Predicting Customer's Loan decision by Logistic Regression

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Introduction

This report studies the factors influencing personal loan acceptance using a logistic regression model. The model is performed on a random sample of 500 observations drawn from the *bankloan.csv* dataset. The binary outcome variable is *personal_loan* (Yes/No), and key predictors include *Income*, *Family*, *Education*, *CD Account*, and *CreditCard* usage. The objective is to identify customer attributes that significantly influence loan uptake, enabling data-driven targeting and credit risk assessment in banking.

Data Preparation

A fixed random seed ensured reproducibility during sampling. Missing data were excluded, and categorical variables such as *Education*, *CD Account*, and *CreditCard* were converted to factor type. The target variable was relabeled for interpretability. Only predictors relevant to the loan decision were retained.

Data Visualization

Figure 1 demonstrates that customers who accepted personal loans exhibited significantly higher income levels compared to those who did not, indicating income as a key predictive factor.

A larger proportion of customers with CD accounts accepted loans as illustrated in *Figure 2*, suggesting a strong relationship between existing product engagement and loan uptake.

Figure 1: Income by Loan Status

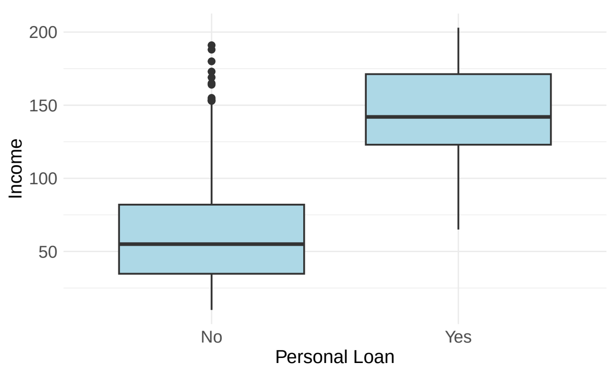

Figure 2: CD Account Ownership by Loan Status

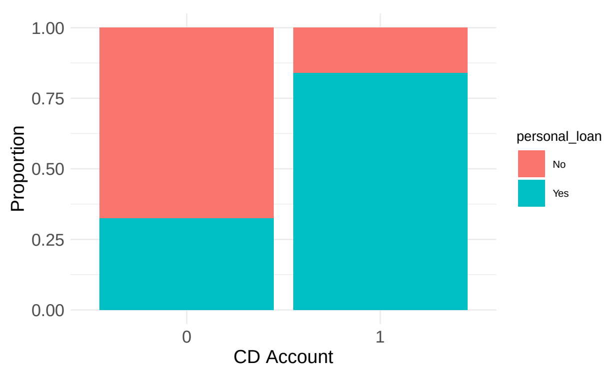

Table 1: Mean Income by Loan Status and CD Account

personal_loan	CD Account	Mean_Income
No	0	63.45583
No	1	76.76923
Yes	0	145.53676
Yes	1	144.32353

Table 1 confirmed that mean income among acceptors was nearly double that of non-acceptors, regardless of CD account ownership, highlighting both variables as key influencers.

Logistic Regression Model

A logistic regression model was considered due to the binary nature of the outcome. The model included five predictors: *Income*, *Family*, *Education*, *CD Account*, and *CreditCard*.

Table 2 presents the regression coefficients, standard errors, test statistics, and p-values for each predictor.

Table 2: Logistic Regression Coefficients

term	estimate	std.error	statistic	p.value
(Intercept)	-10.4077353	1.3335331	-7.8046322	0.0000000
Income	0.0604600	0.0070871	8.5310361	0.0000000
Family	0.6023295	0.2021951	2.9789520	0.0028924
Education2	2.6244958	0.5553586	4.7257676	0.0000023
Education3	2.6355290	0.5701592	4.6224442	0.0000038
‘CD Account‘1	3.2249072	0.7658538	4.2108654	0.0000254
CreditCard1	-0.2023015	0.5147526	-0.3930073	0.6943141

As shown in Table 2, *Income* ($\beta = 0.0605$, $p < .001$) and *Family* size ($\beta = 0.6024$, $p = .0029$) had significant positive effects, indicating higher chances of loan acceptance among wealthier individuals and larger families. Education levels also mattered: those with a higher education were significantly more likely to accept loans ($p < .001$). *CD Account* ownership had the strongest effect ($\beta = 3.22$, $p < .001$), reflecting deeper financial engagement. *CreditCard* ownership showed a negative but statistically insignificant effect on loan acceptance. However, from a behavioral perspective, customers who actively use credit cards may exhibit a borrowing nature and thus be more inclined towards loans. The model's intercept ($\beta = –10.41$) represents the low probability when all predictors are at the base level.

Model Performance

The logistic regression model was applied on the 30% test set to assess how well it predicts loan acceptance.

Table 3: Confusion Matrix

	No	Yes
No	85	9
Yes	6	50

The confusion matrix results shown in *Table 3* highlight strong classification performance. The model correctly identified 50 loan acceptors (true positives) and 85 non-acceptors (true negatives).

The model correctly classified 90% of all cases, indicating high overall predictive performance. Sensitivity (Recall) = 0.847: About 84.7% of actual loan acceptors were correctly identified. Specificity = 0.934: Around 93.4% of non-acceptors were accurately predicted, indicating very low false positives (ref: *Table 4*).

Table 4: Model Performance Metrics

.metric	.estimator	.estimate
accuracy	binary	0.9000000
sensitivity	binary	0.8474576
specificity	binary	0.9340659
auc	binary	0.9620041

Together, these results confirm that the logistic regression model is highly effective and reliable for predicting personal loan acceptance in this dataset.

The ROC curve, as shown in *Figure 3,* indicates excellent separation between the two outcome classes.

Figure 3: ROC Curve for Logistic Regression Model

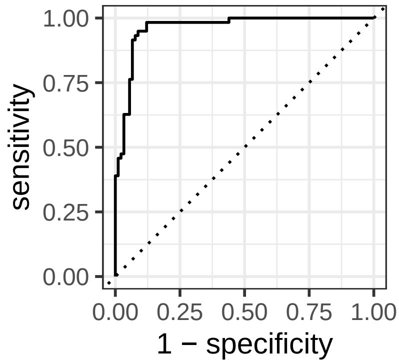

Figure 2

Conclusion

The logistic regression model displays strong predictive performance (90% accuracy, AUC = 0.96). Key predictors such as *Income, Education, Family* size, and *CD Account* ownership were significantly associated with higher loan acceptance likelihood. Based on this model banks can improve targeting by focusing on financially stable, larger families, well-educated customers already engaged with their products, thereby enhancing marketing efficiency and minimizing credit risk.